Check your understanding

1. The graph you have drawn should have six curves on it.

   a. Describe the general trend of all the curves on your graph. [2 marks]

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   b. How many layers would be the most appropriate to keep a beaker of water warm for longest? Use data from your results to support your answer. [3 marks]

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2. Explain why the same volume of water was used each time. [2 marks]

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3. A student wants to keep ice cream frozen when the ice cream is outside the freezer.
   How many layers of insulation you would advise the student to use?
   Explain why you would advise this many layers. [3 marks]

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Exam-style questions

1. **Table 2** shows a student's results after an experiment investigating the insulating properties of different materials.

Table 2

Thermal insulator	Temperature of the pot (°C)					
	Start	1 hour	2 hours	3 hours	4 hours	5 hours
Cling film	90.0	38.0	26.5	25.5	25.0	25.0
Bubble wrap	90.0	43.5	28.0	25.5	25.5	25.0
Polystyrene	90.0	55.0	43.5	37.5	30.0	26.5
Wood shavings	90.0	52.5	40.0	32.5	28.0	25.5

a. Describe a method the student could have used to collect this data.

You should include the safety precautions the student should take. [6 marks]

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b. Explain why the lowest temperature reached is 25 °C. [1 mark]

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4.1.2.1 Thermal insulation

Check your understanding

1. **a.** All of the curves fall quickly at first and then begin to level out (at room temperature). [1]
   (Alternative answers depending on your graph could be: a steady decrease at first, [1]
   and then less of a decrease.) [1]
   b. Your answer will depend on your own data:
   use correctly chosen data from your graph, (e.g. the difference in temperature at 0 and 20 minutes) for number of layers that would be a good choice; [1]
   then compare the difference in temperature to a number of layers that would be a bad choice [1]
   a comment that there is a point at which using additional layers does not give any further benefit. [1]
2. So that valid comparisons can be made of the different materials. [1]
   Because volume of water is a variable – a larger volume of water would take longer to cool. [1]
3. Advise that the student uses a high number of layers, [1]
   as this would slow thermal energy transfer from the environment to the ice cream,
   a comment that there is a point at which using additional layers does not give any further benefit (no additional reduction of energy transfer). [1]

Exam-style questions

1. **a.** Your answer should include: [6]

   Method

   - Fill a beaker with an amount of water.
   - Use the same amount of water each time.
   - Water should be hot/boiling/at 90 °C.
   - Surround the beaker with one type of insulation.
   - Measure the temperature at the start.
   - Measure the temperature every hour for 5 hours.
   - Repeat for all other insulation types.

   Safety precautions

   - Wear eye protection.
   - Be careful with boiling water.
   - Any other valid point.

   b. because 25 °C is room temperature [1]